

Homework 5

1. d.

$$\begin{aligned} SE(\hat{p}_1 - \hat{p}_0) &= \sqrt{\text{Var}(\hat{p}_1) + \text{Var}(\hat{p}_0)} \\ &= \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}} + \sqrt{\frac{\hat{p}_0(1-\hat{p}_0)}{n_2}} \\ &= \sqrt{\frac{0.29(1-0.29)}{300}} + \sqrt{\frac{0.217(1-0.217)}{300}} \\ &= 0.073 \pm 1.96 \cdot 0.0354 \\ &= 0.0354 + [0.00362, 0.142] \end{aligned}$$

2. c. $SE(\hat{p}_1 - \hat{p}_2)$ under the null

$$\begin{aligned} &= \sqrt{\hat{p}(1-\hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} \\ &= \sqrt{0.253(1-0.253) \left(\frac{1}{300} + \frac{1}{300} \right)} \\ &= 0.0355 \end{aligned}$$

d.

$$Z = \frac{\hat{p}_1 - \hat{p}_0}{SE(\hat{p}_1 - \hat{p}_0)} = \frac{0.073}{0.0355} = 2.056$$

3. a. Expected counts under independence

$$\text{Survived w/ Immunotherapy} = \frac{300 \cdot 152}{600} = 76$$

$$\text{Died w/ Immunotherapy} = \frac{300 \cdot 448}{600} = 224$$

$$\text{Survived w/ standard care} = \frac{300 \cdot 152}{600} = 76$$

$$\text{Died w/ standard care} = \frac{300 \cdot 448}{600} = 224$$

$$3. c. \chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(87-76)^2}{76} + \frac{(65-76)^2}{76} + \frac{(213-224)^2}{224} + \frac{(235-224)^2}{224}$$

$$S_i = X^T B + \epsilon_i \quad = 4.26$$

Then χ^2 is the χ^2 (4 d.f.) = 0

So χ^2 is χ^2 = 1

$$P(Y=1) = \frac{e^{X^T \beta}}{1 + e^{X^T \beta}}$$

- this is just two-category case, you can expand to an arbitrary K general. Integrated as the change in the odds of being in a higher category for each unit increase in the covariate shifts equals for each unit of X (proportional odds assumption)

$$\log\left(\frac{P(Y=1)}{P(Y=0)}\right) = \beta - X^T \beta$$

"kind of like a logit

log for each cat. pair

with 1st. max intercept

ODPs for different categories can be subtracted

Even can allow to get the probability of a

certain category